

Applying FAIR Principles to Community-Curated Gaming Data: A Study in Taxonomy, Folksonomy, and Data Curation for Librarianship

Kelsey Kiantoro | INFO628 Data Librarianship and Management Taught by Professor Vicky Rampin | May 2026

INTRODUCTION

Steam's tag system functions as a folksonomy, with 450 tags assigned by players and weighted by votes, lacking a curator and hierarchy. Without intervention, it resembles the network above: indie titles dominate, themes and moods outweigh genres, and terms like *Cozy*, *Souls-like*, and *Wholesome* sit alongside *RPG* and *Shooter* as if they refer to the same thing. They do not.

This project addresses this issue. When community-curated tags are viewed as a flat list, they obscure the structure they carry: what a game *is* (genre), what it *depicts* (theme), how it *functions* (feature), and how it *appears* (visual) are collapsed into a single undifferentiated vocabulary. This makes the data difficult to query, incompatible with controlled vocabularies, and not reusable in accordance with FAIR principles.

Following Grelier, Kaufmann & Schmalz (2023), and validated by Steam's internal Tag Wizard, which organizes tags into these categories, this project employs the same categorical structure internally. Each tag is classified by its description, organized into a three-level genre hierarchy, and linked through a normalized relational database using SKOS properties. The resulting ERD does not replicate the folksonomy; instead, it applies documented classification decisions to it.

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METHODOLOGY

1. From Flat Data to Relational Structure

The project began with two datasets. The primary dataset was;

- The [Top 1,000 Steam Games \(2024–2026\)](#) by Waddah Ali (Kaggle, CC BY-NC-SA 4.0), 1,000 game records scraped from the Steam API and SteamSpy in March 2026.
- The secondary dataset was [a Gamalytics publisher list processed by Seyed](#), covering 205 Steam publishers with country, revenue estimates, and market-tier classification.
- An earlier tag vocabulary source ([Steam Trends 2023 by Barabanov & Kobelev](#)) provided the initial 445-tag pool used to build the taxonomy framework.

The publisher dataset was cleaned and included in the database schema (PUBLISHER table, ERD), but it was not linked to the taxonomy and folksonomy analyses. Since the project scope focused on tag classification and controlled vocabulary comparison. Both data files were flat, with multiple values per row, no hierarchy, inconsistent use of controlled vocabulary, and no referential structure.

The first analytical step was transforming this flatness into a normalized relational database. The normalization process (1NF → 3NF) revealed that one game could have many tags, one publisher could release many games, and one tag could appear across thousands of games, none of which a flat CSV can express cleanly. This led to the design of a 7-table relational schema:

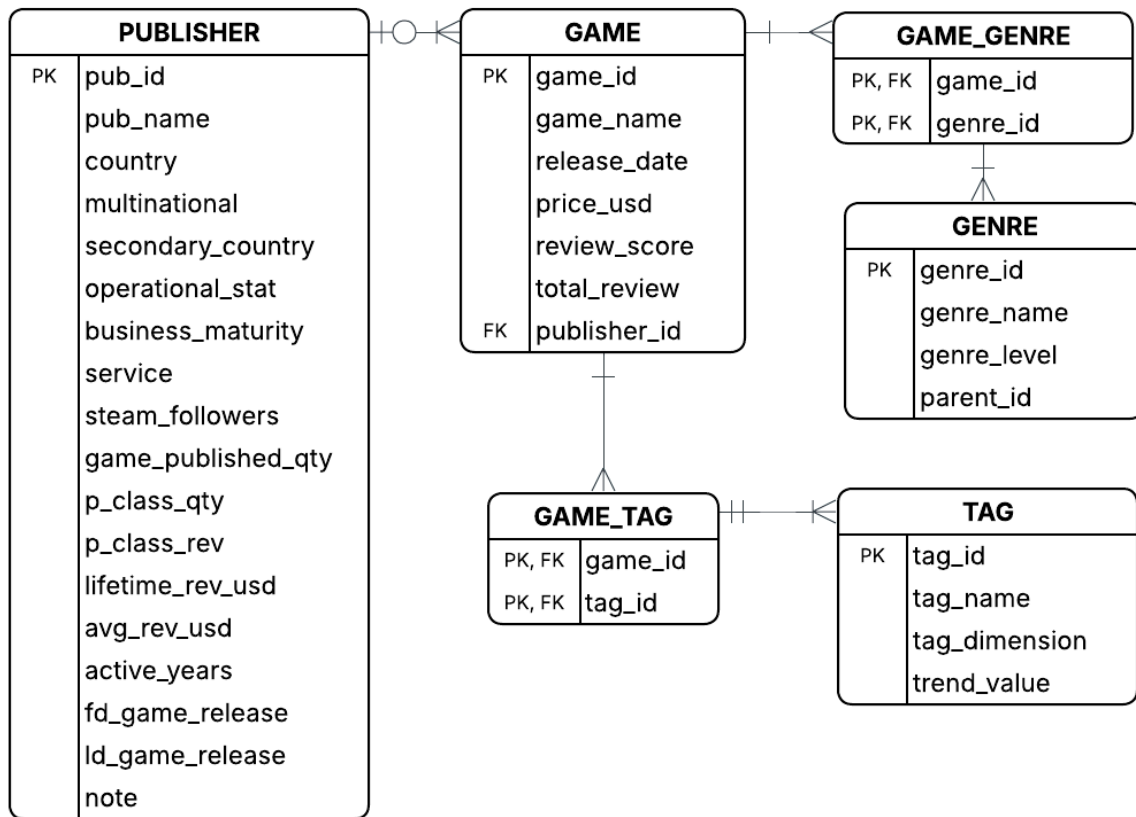


Figure 1. The developed ERD map captures both genre and tag in the game category that satisfied the 3NF Framework, 2026. Source: Author.

Three data quality challenges were resolved during this phase.

1. First, publisher name normalization: variants such as 'EA,' 'Electronic Arts,' and 'EA Games' were clustered using OpenRefine and mapped to a single canonical name.
2. Second, type coercion with error handling: numeric fields (price, review score, peak players) were converted using try/except logic with None fallback to preserve records rather than drop them.
3. Third, genre string reconciliation: the raw data used inconsistent spellings; 'Co-op,' 'Do-Op,' 'Coop,' which were normalized using a mapping

2. Genre Classification: Building the Taxonomy

Steam's tagging system is a folksonomy; players vote on tags for each game, and any tag can be applied to any game with no editorial control, no hierarchy, and no dimension labels. The raw dataset contained 445 unique tags, and all of them were treated as equal: 'RPG,' 'Cyberpunk,' 'Co-op,' and 'Pixel Art' sat in the same flat list despite describing fundamentally different aspects of a game (what it is vs. what it looks like vs. how it is played vs. what it depicts).

The classification process followed Grelier, Kaufmann, and Schmalz's (2023) framework, which identifies that Steam tags span multiple non-equivalent dimensions.

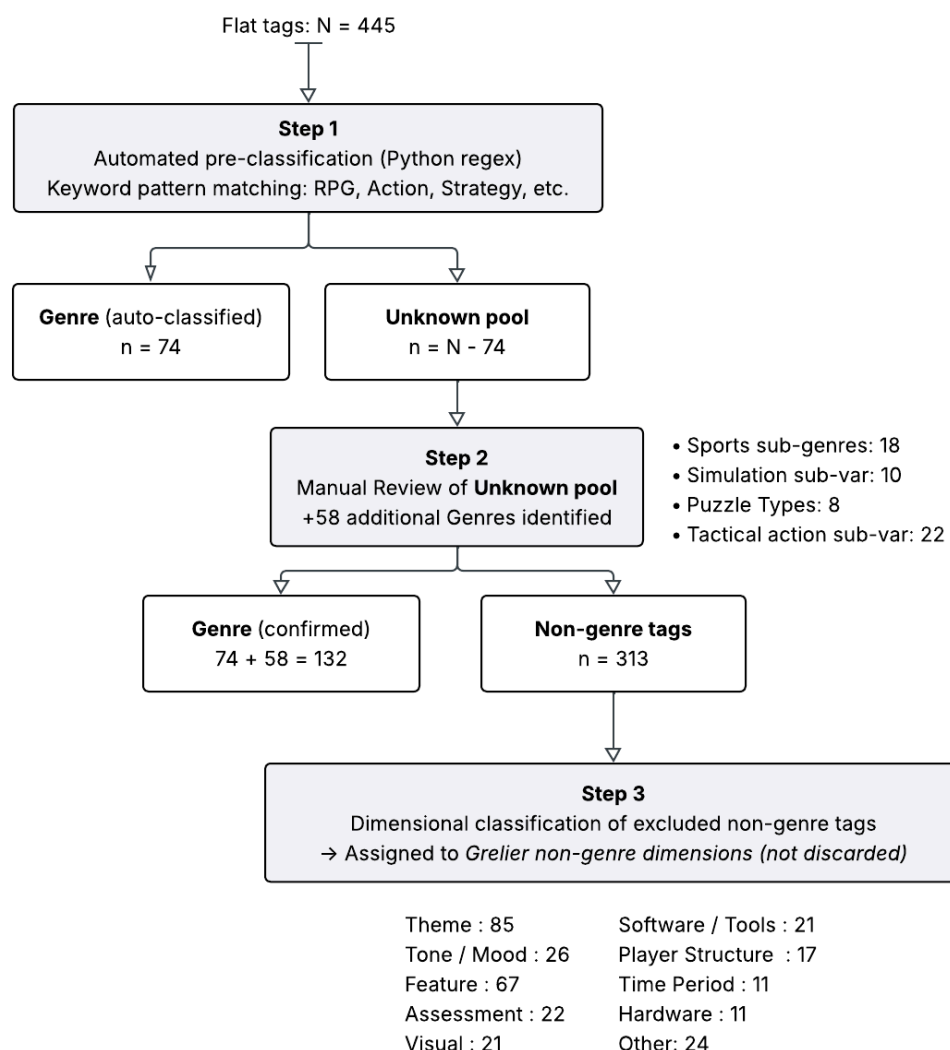


Figure 2. The processing steps for transforming the tag hierarchy from flat to nested using the *Steam Trends 2023* (Barabanov & Kobelev) and *Top 1000 Steam Games 2024–2026* (Waddah Ali, Kaggle, CC BY-NC-SA 4.0). Note. Total 3-tiered steps. Source: Author.

A three-step pipeline was applied:

Step 1: Automated pre-classification (Python regex): 74 tags were auto-classified as Genre based on keyword pattern matching (RPG, Action, Strategy, Shooter, Platformer, etc.). This established a starting floor.

Step 2: Manual review: 58 additional genres were identified in the 'unknown' pool, including sports sub-genres (18 tags), simulation variants (10 tags), puzzle types (8 tags), and tactical action sub-types (22 tags). Total confirmed: 132 genres.

Step 3: Dimensional classification of excluded tags: The remaining 313 non-genre tags were not discarded. Each was assigned to one of Grelier's non-genre dimensions — Theme (85), Feature (67), Tone/Mood (26), Assessment (22), Visual (21), Software/Tools (21), Player Structure (17), Time Period (11), Hardware (11), Business Model (7), and Content Rating (6).

The resulting taxonomy was organized into three hierarchical levels: **9 Super-Genres** (Action, RPG, Adventure, Puzzle, Strategy, Simulation, Sports, Racing, Casual), 78 mid-level Genres, and 41 Sub-Genres, totaling 128 hierarchical entries. Edge cases were resolved explicitly: Action RPG was assigned to RPG (character progression dominant), Puzzle Platformer was assigned to Puzzle (puzzle-solving is the primary mechanic), and Survival was excluded as Theme (describes setting, not core mechanic, per Grelier et al.).

ANALYSIS

Finding 1# Relationship between tags

The central analytical finding emerged from comparing the structured **SteamDB** taxonomy against **MobyGames'** controlled vocabulary. **SteamDB** operates as a pure folksonomy: tags emerge bottom-up from millions of player votes, with no editorial control or hierarchy. The network analysis aimed to help determine the hierarchy base.

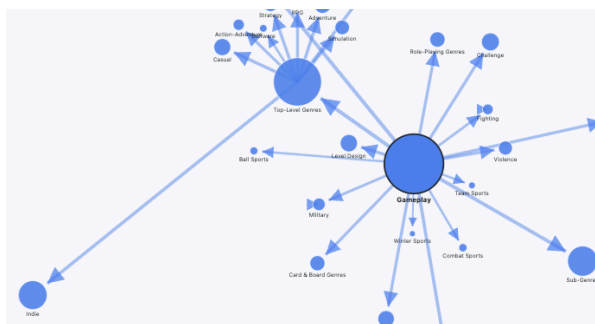


Figure 3. SteamDB Genre Network Graph. Note: The Indie category is the largest under the Top-level genre umbrella in 2026. Source: author-processed from SteamDB categories in 2026, assisted by Claude.

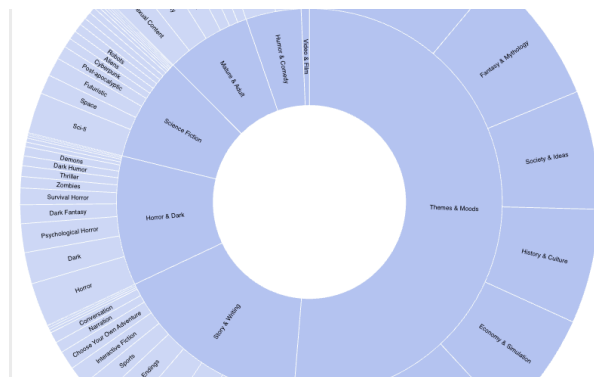


Figure 4. SteamDB Genre distribution Sunburst Diagram, 2026. Source: <https://observablehq.com/d/3ebcdd430b611277?collection=@kelsey-kiano/steamdb-genre-analysis>

More significantly, *Themes & Moods* is the largest cluster overall when measured by cumulative tag weight, larger than the Genre category itself. Players spend more vocabulary describing how a game feels (Atmospheric, Cozy, Dark, Wholesome) than what it mechanically is. This is the clearest evidence that Steam's folksonomy is an affective system, not a classificatory one,

and the core reason it cannot be treated as a controlled vocabulary without the dimensional separation applied here.

Note: this network structure reflects SteamDB's raw tag-weight distribution across the full Steam catalog, and is not derived from the GENRE or TAG tables in this project's database. The relationship analysis was conducted to help develop a nested relationship for the genre-tag categorization in the ERD.

Finding 2# From Flat Data to Relational Structure

By comparing the structured Steam taxonomy with MobyGames' controlled vocabulary, we observe a significant reduction in the dataset's structure. MobyGames functions as a library-style classification system: editors assign controlled genre terms from a predefined vocabulary using a top-down, authoritative approach, suitable only for validation. Steam operates as a pure folksonomy: tags emerge bottom-up from millions of player votes, with no editorial control or hierarchy.

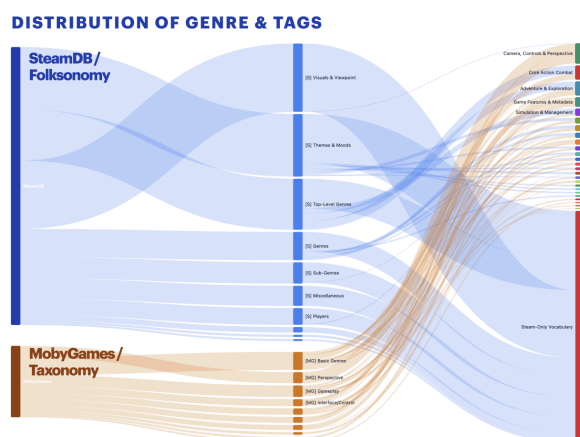


Figure 4. Distribution of genre and tags from SteamDB to MobyGames. Only about 15% of the tags were considered genres. Source: author processed from SteamDB and MobyGames categories in 2026.



Figure 5. Boomer Shooter tag homepage from Steam, 2026. Note: "Boomer shooters" (or retro FPS games) are a massively popular subgenre of first-person shooters that channel the fast-paced, action-oriented, and high-mobility spirit of 1990s classics like DOOM and Quake. Source: <https://store.steampowered.com/tags/de/Boomer-Shooter/>

The 386 Steam-exclusive tags represent the most analytically significant gap. These are terms that the gaming community uses naturally, such as 'Cozy,' 'Boomer Shooter,' 'Wholesome,' 'Vibe,' 'Souls-like,' that no cataloger would invent but that describe real player experiences. MobyGames, by contrast, captures structural metadata that players rarely use: 'Perspective,' 'Visual Presentation,' 'Pacing.' The Stardew Valley case study illustrates this precisely: Steam

users tag it with 'Farming Sim,' 'Relaxing,' and 'Agriculture,' while MobyGames classifies it under 'Life Sim' and 'Simulation' with no emotional or experiential dimension.

The network graph analysis confirmed that Indie is the single largest Top-Level Genre node in the Steam taxonomy (129,907 games), bigger than Action, Casual, and Adventure combined. Games and Moods is the largest cluster overall (786,602 cumulative tag weight). These patterns are invisible in MobyGames' controlled vocabulary, not because they are wrong, but because the two systems answer different questions and try to serve different purposes.

Finding 3# Who Validates the Correct Tag?

A secondary but important finding concerns the epistemology of validation. In MobyGames, an editor decides what genre a game belongs to. In Steam, the community votes, but voting weight alone does not guarantee accuracy. A game with 786,602 tag applications could simply reflect marketing success or community trends rather than inherent genre membership.

This raised a practical question in data librarianship: when two authoritative sources disagree, which takes precedence? Examining both simultaneously is more informative than choosing one. For example, MobyGames classifies many titles under the broad 'Action' category, while Steam communities have fragmented that category into Hack and Slash, Soulslike, Character Action Game, Spectacle Fighter, and Musou, distinctions that reflect how players actually experience and discuss the games.

The absence of certain Steam tags from MobyGames is equally revealing. Tags like Gore, Sexual Content, and NSFW appear frequently in Steam's community vocabulary but are absent from MobyGames' genre taxonomy; the assumption is that a controlled vocabulary system makes explicit editorial choices about what categories to expose. MobyGames handles content descriptors through a separate attribute system rather than treating them as genre equivalents. This is itself a classification decision, reflecting institutional values rather than community consensus.

MANIFESTO

SKOS as the Bridge

This led to a two-table design, inspired by the W3C SKOS (Simple Knowledge Organization System) standard for representing controlled vocabularies and folksonomies within a single structure (Miles & Bechhofer, 2009).

The GENRE table encodes the hierarchical taxonomy: genre_id, genre_name, genre_level (ENUM: Super-Genre, Genre, Sub-Genre), and a self-referencing parent_id foreign key — where the parent-child relationship maps directly to skos: broader. Grelier et al.'s dimensional framework was selected in part because it aligns with Steam's own internal Tag Wizard (Valve Corporation, n.d.), which organizes tags into the same categories: Genres (Super-Genre → Genre → Sub-Genre), Visuals & Viewpoints, Themes & Moods, and Features. The classification applied here does not impose an external library framework onto community data; it recovers a structure that Steam itself uses internally but does not expose in its public interface.

The TAG table encodes all non-genre community tags: tag_id, tag_name, tag_dimension (ENUM: Theme, Feature, Tone, Visual, Time_Period, Hardware, Assessment, Player_Structure, Content_Rating), and trend_value, where each tag maps to skos:inScheme, indicating membership in the controlled taxonomy without imposing a genre hierarchy on it.

GENRE			
PK	genre_id	SMALLINT	
	genre_name	VARCHAR (100)	
	genre_level	ENUM (Super-Genre, Genre, Sub-Genre)	← skos:broader
FK	parent_id	SMALLINT	

TAG			
PK	tag_id	SMALLINT	
	tag_name	VARCHAR (100)	
	tag_dimension	ENUM (Theme, Feature, Tone, Visual, Time_Period, Hardware, Assessment, Player_Structure, Content_Rating)	← skos:inScheme
	trend_value	ENUM (High, Medium, Low) NULL allowed	

Figure 6. Genre and Tag entity's Constraint and value. Note, SKOS applied. Source: Author.

Importantly, the GENRE hierarchy is mereological rather than strictly taxonomic, meaning Super-Genres function as containers that group related genres by family resemblance (Li & Zhang, 2020), rather than as parent classes from which child genres inherit properties. This is encoded using `skos:broader`, which in SKOS practice covers both IS-A and part-whole relationships, though the intended relationship here is the latter.

Together, the two tables hold both the structured taxonomy and the flat folksonomy within one schema, directly satisfying the FAIR principle of Reusability (Wilkinson et al., 2016): the dataset becomes usable by researchers who need controlled genre hierarchies and by developers who need the full community tag ecosystem, without forcing either to adopt the other's language.

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RELATED SOURCES

- Data Management Plan
- Code, Database (Work in progress)
 - Visualization: Network Graph
 - Visualization: Sunburst and Sankey Diagram
- Research Poster

